Chapter 7

Cores

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INTRODUCTION

A core material replaces missing coronal tooth structure prior to restoration with an indirect extracoronal restoration and helps to stabilize weakened parts of the tooth. It is generally recommended that a core should be considered when more than 50% of the coronal part of the tooth is missing. Existing restorations should be investigated radiographically to diagnose residual caries or caries adjacent to the restoration and evaluated clinically to determine if the existing restoration needs replacing. An assessment of the depth of the cervical margins should highlight potential challenges with preparation and impression taking. Some clinicians recommend the routine removal of existing directly placed restorations to confirm these criteria and replacement with a new core to ensure that the newly placed core is adequately retained (see Chapters 3 and 8 for further discussion). But this is not always necessary, particularly if the restoration was placed recently or there are no clinical or radiographic reasons to do so.

MATERIALS

Several dental materials can be used for core build-ups; each has different properties and therefore advantages and disadvantages. Enamel–dentine bonding using adhesive materials, such as composite resins or glass ionomer-based materials, allows a more conservative technique

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compared to amalgam which often requires additional tooth preparation to achieve adequate retention. Crown preparation can be carried out immediately following core build-up if the material used is command set by light curing. If the core is placed as a transitional restoration at an appointment prior to preparation then it should be sufficiently contoured to give occlusal stability, cause no food packing and maintain gingival health.

There are many materials on the market that are suitable for core build-up, but those based upon composite resin are increasing in use and many may say they are the most appropriate to use in contemporary practice. The use of amalgam has decreased in popularity, mainly because of health and environmental issues over its mercury content, its inability to bond to tooth structure and its colour. In a number of countries the use of dental amalgam has already been banned and this is only likely to extend to other countries in the future. However, it is still in use in sufficient countries and sufficient quantities to warrant its description in this textbook.

Composite resin (Figure 7.1)

Composites consist of a resin, normally an aromatic dimethacrylate such as BisGMA, and filler particles such as quartz, silica and other types of glass. The type, particle size and content of the filler particles control the properties of the material. Barium- or strontium-containing glasses are also added to make the material radio-opaque. Many composites have a variation in particle size. So-called hybrid composites contain large filler particles (15–20 μm) and smaller colloidal silica (0.01–0.05 μm) particles. Microfilled composites (average filler particle size = 0.02 μm) can be highly polished, but this is a property obviously not necessary for a core material. The most effective core materials are hybrid composites and more recently the newly developed low-shrink composites (e.g. Filtek LS Low Shrink Composite (3M ESPE) based on silorane chemistry) might prove to be effective. Composites have a compressive strength similar to dentine and higher tensile and flexural strengths. They can be bonded to tooth structure when used with a dentine-bonding agent; however, placement is technique sensitive.

When using composites it is important to prevent moisture contamination with saliva or blood and, therefore, where possible, rubber dam isolation is recommended. Visible light cured composites should be placed in increments to reduce problems with shrinkage on setting. Polymerization contraction may increase the risk of marginal leakage and post-treatment sensitivity. The lower shrink composites that are being introduced may overcome this problem in the future but will need assessing as the materials develop. Water sorption occurs with composites and can lead to expansion of the material. This, however, takes some time to occur and is unlikely to have an impact on the provision of indirect restorations. Any dimensional change that does occur will be easily compensated for by the placement of die relief on the master die in the laboratory. Composite is the material of choice for a core when an all-ceramic crown is planned (Figure 7.1).



Figure 7.1 Composite resin cores placed in root-filled teeth in the upper right sextant in preparation for all ceramic dentine-bonded crowns. See Chapter 11 for preparations and cemented restorations.

Newer hybrid composite core materials are available with various additives such as fibres, ceramic fillers, titanium and lanthanide, that claim to improve the mechanical properties of the material. Examples of these are Paracore (Coltène Whaledent), Ti-core (Essential Dental Systems, Inc.), Light-Core (Bisco, Inc.), Coradent (Vivadent) and Core Paste XP (Den-Mat). Composite core materials are available with a blue pigment to allow the clinician to distinguish between tooth structure and material during preparation (e.g. MultiCore Flow Blue, Ivoclar). Some composites are also packable, facilitating contact point formation and reducing the risk of voids.

Advantages of composite resin as a core material

- Strong and can therefore be placed in thinner sections compared to amalgam
- Immediate setting with light-cured composites
- Can be bonded to tooth structure
- No mercury
- Tooth coloured, hence ideal under all ceramic crowns

Disadvantages of composite resin as a core material

- Technique sensitive – moisture contamination and polymerization shrinkage should be avoided
- Can be difficult to distinguish between composite and tooth structure when preparing crown margins

Glass ionomer cement (GIC)

The original cements contain a fluoroaluminosilicate glass, which reacts with a polyalkenoic acid to form a cement. Many studies have shown that glass ionomer is not sufficiently strong to be used as a core build-up material unless there are two intact walls remaining. There should also be at least 1–2 mm of remaining sound tooth structure coronally that can be prepared as a ferrule. This is an important distinction when compared to composite materials that have greater strength and are more suitable for extensive restorations. Glass ionomer can also be useful as a filler to block out undercuts (e.g. when preparing a tooth for an inlay) and to make good any defects or irregularities in a tooth preparation for an indirect restoration (e.g. when a filling or piece of tooth is lost during tooth preparation; see Chapter 12). Variations of glass ionomer cements have been developed in the past in an attempt to improve the physical properties of the material and these have included the addition of metal particles by a fusion process resulting in a cermet, or addition of amalgam alloy particles resulting in an admix.

Resin-modified glass ionomers (RMGIs)

In the early 1990s a water-soluble resin (hydroxyethyl methacrylate, HEMA) was added to conventional GIC to improve the physical properties, as such the material cures by an acid–base reaction (the glass ionomer component) and a resin polymerization, which is either chemical or light activated or both (dual cured). Some RMGIs are specifically advocated as core materials such as Vitremer core build-up material (3M ESPE). Whilst most of these materials are tooth coloured, their use beneath all-ceramic restorations should be avoided as they undergo hygroscopic expansion which could lead to ceramic fracture. Resin-modified glass ionomer is an ideal core material for metal or metal–ceramic crowns, inlays, onlays or bridges, or reinforced core ceramics; however, like chemically cured glass ionomers, it needs significant remaining tooth structure to be effective.

Advantages of GIC and RMGIs as core materials

- Sets quickly, allowing immediate preparation
- Adhesive
- Fluoride release – The clinical significance of this is controversial and there appears to be no conclusive evidence in the literature for or against the anti-cariogenic properties of fluoride release from GICs or RMGIs in clinical trials (Randall & Wilson, 1999)
- Low thermal expansion coefficient

Disadvantages of GIC and RMGIs as core materials

- Low compressive and tensile strengths
- Weak material only suitable when significant proportion of the tooth remains
- Deterioration at low pH
- Sensitivity to moisture during setting

Amalgam (Figure 7.2)

Dental amalgam consists of mercury combined with a powdered silver–tin alloy with the addition of copper, palladium and other elements. High copper amalgam



Figure 7.2 Failed metal–ceramic crowns in the upper right sextant being removed (top) and the amalgam cores beneath (bottom).

alloys have a copper content of 30% and on setting have a smaller concentration of the gamma-2 phase, which means they are less easily deformed, stronger in compression, and have reduced potential for corrosion. The alloy particles can be spherical, lathe cut or admixed. Amalgam can be the material of choice for core build-ups on posterior teeth and is still used by some practitioners. Its use is, however, on the decline and is only likely to decline further in the future. Unfortunately, as the material takes 24 hours to reach maximum compressive strength, crown preparation has to be carried out at a second appointment. The spherical, high copper alloys reach their maximum strength faster and can be prepared 10–15 minutes following placement rather than the normal 24 hours; however, even this is a long time to wait in a busy practice. As amalgam does not adhere to tooth structure it has to be mechanically retained or chemically bonded. It performs well if properly condensed and is not contaminated by large amounts of blood or saliva during placement.

Advantages of amalgam as a core material

- Not particularly technique sensitive
- Strong if placed in sufficient bulk
- Can be used as a bonded amalgam
- Easy to distinguish between amalgam and tooth structure
- Packable – if condensed well voids are avoided

Disadvantages of amalgam as a core material

- Long setting time
- Low initial tensile/compressive strength
- Weak in thin section
- Mercury may be of concern to some patients
- Potential for electrolytic galvanic action between amalgam and other metals in crowns for example
- Not adhesive and so needs mechanical retention derived from the cavity

VITAL TEETH

Preoperative assessment (see also Chapter 3 on endodontology)

It is important prior to planning an indirect restoration that the tooth and any existing restoration are carefully assessed to ensure long-term success. The tooth should be symptom free, ideally provide a positive sensitivity test and a periapical radiographic examination made to ensure an absence of periradicular pathology. If needed, endodontic treatment should be carried out. The existing restoration needs careful examination and if doubt exists it should be removed to ensure that the core is placed on sound tooth structure and that no previous carious exposure exists.

Methods of core retention for vital teeth

There are no absolute guidelines, but if more than 50% of tooth structure has been lost or removed, additional methods of retention are usually required. Retention can be either mechanical, when using amalgam, or adhesive. The latter is the most common as composite is the preferred material for a core.

Adhesive retention

Composite resins and dentine bonding

Bonding of composite to etched enamel has made a huge advance in operative dentistry, allowing aesthetic restorations to be placed and clinical techniques such as minimum preparation (resin retained) bridges. Bonding of hydrophobic composite resins to physiologically wet dentine has also made a massive leap forward over the last two decades. Generally the dentine surface is chemically treated with an acid to allow the mechanical interlocking of resin around the dentinal collagen, producing a layer known as the hybrid layer. The bonding agent, which is normally an unfilled resin, is then applied to the dentine surface and light cured. It co-polymerizes with the resin already present in the collagen matrix, locking it onto the dentine and providing a more suitable surface for bonding with resin composite materials.

The steps involved in dentine bonding have changed over the years. Initially a separate etch, primer and bond were used. With this technique there was a risk that during drying the collagen may collapse and ruin formation of the hybrid layer. Two-step systems have either a separate etch with the prime and bond combined in one bottle or the etch and prime combined with a separate bond. The latter has the advantage that the self-etching priming agent does not have to be washed off the dentine. One-step bonding systems are similar to self-etching priming materials but have the bonding agent added also. They have been shown not to etch the enamel as effectively as phosphoric acid.

Adhesive retention for amalgam restorations – amalgam bonding

Whilst dentine bonding was developed for use with composite, its use has also been applied to the more traditionally used amalgam. Amalgam-bonded restorations are thought to improve restoration retention, reinforce remaining tooth tissue and enhance the marginal seal against bacterial leakage. This philosophy is not new; in 1897, Baldwin suggested placing a thin, wet layer of zinc phosphate cement on the cavity walls prior to condensing the amalgam in an attempt to improve the bond and marginal seal; the acidity of the unset zinc phosphate cement

probably etched the tooth, creating micromechanical retention. Today, the technique has been modified using modern dental materials including self- or dual-curing metal adhesive resins or glass ionomer cements. Specific products such as All-Bond 2 (Bisco), Amalgambond Plus (Parkell), Optibond 2 (Kerr), RelyX ARC (3M ESPE) and Panavia EX, Panavia 21 (Kuraray) have been studied regarding bond strengths and prevention of microleakage.

Amalgam bonding – technique

- Gain as much mechanical retention as possible using conventional means, such as undercuts, occlusal key ways and grooves.
- Isolate the tooth using rubber dam or cotton wool rolls and saliva ejector.
- Consider if a lining is needed in a deep cavity, keeping this to a minimum, bearing in mind this might interfere with the subsequent bonding technique.
- Check the fit of matrix band, remove and lightly coat its inner surface with petroleum jelly.
- Isolate the cavity but do not over-dry as this may result in postoperative sensitivity. Place the matrix band using wedges to maintain a tight fit gingivally.
- Either use a total etch technique in the cavity, rinse thoroughly, dry and apply a compatible dentine-bonding agent according to the manufacturer's instructions (e.g. Scotchbond for use with RelyX ARC) or use a compatible self-etching bonding agent.
- Using a microbrush, apply a thin coat of freshly mixed resin (e.g. RelyX ARC) over the entire cavity surface.
- Condense the amalgam onto the unset resin, beginning in the deepest areas such as grooves and boxes first. The wet cement will be extruded towards the occlusal surface.
- Once the cavity has been completely packed, remove the top layer of amalgam/cement with a cotton pellet and replace with a final layer of amalgam. Complete initial carving before removal of the matrix band.

Mechanical retention for amalgam restorations

Intradental pins

The use of dentine pins has decreased as the use of amalgam has declined (and hence its need). Pin placement is also fraught with problems, but, like amalgam, some practitioners continue pin use so it is sufficient to warrant description in this text. Most intradental pins are self-threading; a pin hole is cut within dentine which is slightly narrower in diameter than the thread on the corresponding pin (Figure 7.3). Therefore, as the pin is inserted into the hole, it cuts its own counter-thread on the walls of the pin hole, causing stresses within the dentine (Figure 7.4). As the pin is inserted deeper into the hole, the frictional forces increase and often the pins shear off before they reach the base of the pin hole (Figure 7.5). This leaves a void at the base of the pin hole, an unretentive and potentially loose pin in dentine, and a pin too long coronally which is likely to interfere with the occlusion. Any attempt at shortening the pin with a high speed bur will cause vibration of the pin, shattering the dentine counter-threads, and will lead to loss of the pin. Clamping the pin with a self-locking tweezers will dampen down the vibration and will reduce the risk of loosening the pin. Alternatively, the pins can be bent slightly to avoid the occlusion and to ensure that they are surrounded by sufficient core material (Figure 7.6).

The use of pins to aid retention should only be employed as a last resort due to their complications. In addition, if placed in the wrong direction, they can completely or partially perforate into the pulp or the



Figure 7.3 Self-threading intradental pins. The pin drills cut a hole in the dentine slightly narrower in diameter to the thread on the pins. The pins shear off once they have penetrated the full depth of the pin hole.

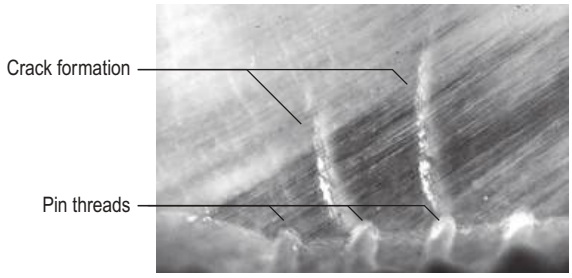


Figure 7.4 Micrograph showing crack formation in dentine radiating out from the pin threads cut into the pin hole seen at the bottom of the image.
(Courtesy of Professor A Grieve)



Figure 7.5 Radiograph of two pins in the upper right first premolar tooth. The mesial pin has not completely entered the dentine and the distal appears to be perforating into the pulp (note this is a two-dimensional image and the pin may be buccal or lingual to the pulp).

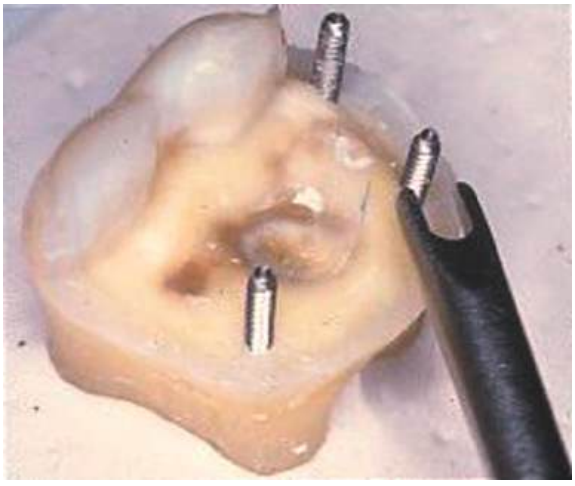


Figure 7.6 Three pins placed in an extracted molar tooth. The pins are tilted and need to be bent to ensure sufficient core material surrounds them.

periodontal ligament (Figures 7.5 and 7.7, respectively). On the few occasions where pins are thought to be necessary the following guidelines should be considered:

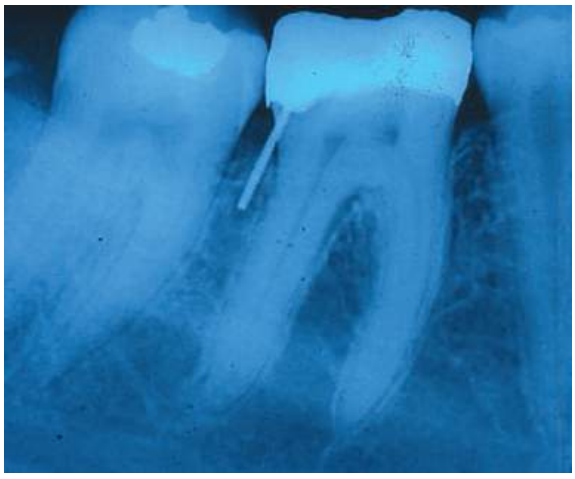


Figure 7.7 The pin placed in this molar tooth has perforated into the periodontium and will be extremely difficult to remove – this may need to be done surgically.

- As pins weaken the amalgam restoration use the minimum number. As a guide, use one pin for every missing cusp or marginal ridge up to a maximum of four. The stresses formed in pin-retained cores can be reduced by coating the pin with an adhesive such as Panavia and 4-META (4-methacryloxyethyl trimellitate anhydride).
- The pin should be placed 2 mm into dentine and 1 mm away from the enamel–dentine junction.
- Pins should be placed parallel to the external tooth surface and away from furcation areas and proximal boxes.

It is important to check the pin channel carefully for any perforations prior to inserting the pin; failure to do so can lead to the need for complicated endodontic treatment or periodontal surgery to remove the pin (Figure 7.7). If preparation of the pin channel causes perforation into the pulp chamber the channel should be filled with mineral trioxide aggregate (MTA). If the perforation is into the gingival sulcus this should be included within the margin of the preparation; however, if below the epithelial attachment the channel may again be filled with MTA.

Circumferential slots and grooves

Introduced by Outhwaite and colleagues (1979), these were originally prepared in dentine using a very small inverted cone bur (a small round bur could also be used) and were found to be equally retentive as the placement of four dentine pins. The core, however, is more easily displaced during removal of the matrix band when amalgam is used. If amalgam is used, consideration should be given to the placement of an orthodontic band (Figure 7.8) or a correctly contoured and polished copper band, which can be left in place whilst the amalgam sets.



Figure 7.8 Orthodontic bands placed on heavily broken-down teeth to stabilize the core and aid their retention to allow endodontic treatment. Such bands can be used when placing amalgam cores and removed once the amalgam has set.

Elective root canal treatment

Teeth with a doubtful pulpal status can be considered for elective root canal treatment prior to provision of an indirect restoration. The success rate of endodontic treatment carried out when teeth are vital with minimally infected pulp tissue is higher than when the pulp has become necrotic and heavily infected, and when pathology of the periradicular tissues has become involved.

Elective endodontic treatment may also be necessary in teeth that have lost substantial coronal structure because of caries, tooth wear, trauma or fracture of weakened tooth tissue around restorations. These teeth may have so little remaining tooth structure that there is insufficient retention for a core and an elective endodontic treatment is required to allow placement of a post (usually in single rooted teeth) or utilization of the pulp chamber (in molar teeth) to help retain the core.

Vital teeth that have overerupted into an opposing edentulous space or where there is an inadequate opposing restoration may require occlusal correction to bring the tooth/teeth back into an acceptable occlusal plane. This may necessitate endodontic treatment as decreasing the occlusal height can expose significant amounts of dentine, causing intractable sensitivity or in some cases may itself expose the pulp (Figure 7.9). In these situations it is important to re-establish a stable occlusion to prevent further overeruption.

Occasionally tooth preparation may result in irreversible pulpitis; this can occur because of a history of repeated restorations, overzealous tooth preparation or insufficient water spray during preparation (see Chapter 3 on the stressed pulp). Patients should be warned of this complication in advance. Often the symptoms do not present until after the final restoration has been fitted, presenting the dentist with the embarrassing dilemma of either carrying out root canal treatment through the newly placed restoration (which is technically much more difficult) or removing the restoration and replacing it following root canal treatment.



Figure 7.9 Patient with an overerupted upper canine (top). To align this tooth it has had to be root treated to enable sufficient reduction in height for alignment and further reduction to allow for crown coverage (bottom). Note the lower premolar tooth has been built up into occlusion to prevent further overeruption.
(Courtesy of Suzanne Blacker)

ENDODONTICALLY TREATED TEETH

It was a traditional belief that endodontic treatment caused teeth to become more brittle and therefore prone to fracture. This was thought to be because of loss of moisture content and changes in collagen cross-linking in dentine. This has, however, been disputed by recent research, which has found that the brittleness occurs as a result of the combination of loss of coronal tooth structure and structural integrity caused by the access preparation. Since tooth fractures are more common in

endodontically treated teeth, the design and prompt placement of the definitive restoration is of huge importance to increase the longevity of the tooth. The type of restoration will depend on the following:

- *The amount of remaining tooth structure*. The loss of tooth structure is one of the most important factors that determine the type of definitive restoration. Whilst endodontically treated teeth with only minimal access cavities can be restored conservatively, those weakened by considerable damage, including loss of one or both marginal ridges, will require more extensive restoration, sometimes involving post placement for core retention and a crown. In all situations, preservation of as much coronal tooth tissue as possible is important as it affords the crown a ferrule effect, protecting the tooth from an increased risk of root fracture.
- *The position of the tooth*. Anterior teeth that are intact other than the endodontic access cavity are at little risk of fracture and generally do not require a crown. An endodontically treated tooth that has lost substantial tooth structure will require a crown and possibly a post if there is insufficient tooth structure to retain a core. Posterior teeth are subjected to greater occlusal loads and must therefore be protected against fracture. Cuspal coverage restorations are generally indicated. Wherever possible the core should be mechanically or adhesively retained and posts avoided if possible.

When to place a post?

Anterior teeth with minimal palatal access cavities require restoration only with a simple composite filling. All traces of gutta-percha and sealer should be removed from the pulp chamber to the level of the amelo-dentinal junction and sealed with a layer of resin-modified glass ionomer to form a coronal seal. A lighter shade of composite is used to maximize the aesthetics of the tooth and for obvious reasons anterior teeth should not be restored with amalgam.

A palatal access cavity often weakens teeth with existing extensive mesial and distal restorations to the point where a crown should be placed as the definitive restoration. If there is sufficient coronal tooth structure to retain a composite core once the tooth has been prepared for a crown, then a post is not essential or required as generally this does not strengthen the tooth/root (Figure 7.10); a post is only indicated when there is insufficient remaining coronal tooth structure to retain a core.

Placing posts in posterior teeth provides more challenges, partly because of the difficulty of access but also because the root canals tend to be narrower. A post is only needed in posterior teeth if there is insufficient coronal tooth tissue remaining to support a core. In most cases a Nayyar core can overcome the need for a post. This core utilizes the shape of the access cavity to retain the restoration (see later in the chapter). If there is little pulp chamber present as a result of extensive tooth tissue loss, a Nayyar core is unsuitable and in this case a post is appropriate. When situations like this arise the overall prognosis for the tooth is generally poor.



Figure 7.10 Periapical radiographs of upper left central incisor and upper left canine. The teeth are bridge abutments. The central has a serrated ParaPost and ~5 mm of apical gutta-percha. The canine did not require a post as there was an adequate amount of coronal tooth tissue remaining.

The most important determinant to the successful restoration of a root-filled tooth is the presence of a ferrule. This is the name for a rim of supragingival tooth tissue which is used to assist retention of the restoration and is explained in more detail later in the chapter.

ANTERIOR AND SINGLE-ROOTED TEETH

Post space preparation

Post space preparation should be carried out with care so as not to disrupt the apical seal or cause a lateral perforation. At least 4–5 mm of gutta-percha should be left to provide an adequate apical seal (Figure 7.10). If this leaves insufficient length of root canal for retention of a post, then the root should be extracted as the restoration will be compromised because of lack of retention. Root canals that have been obturated using a heated carrier technique such as Thermafil make it more difficult to remove only the coronal portion without disrupting the apical part; therefore, if it is obvious that a tooth will require a post, this type of obturation technique should not be used.

Removal of root canal filling prior to preparation of post space

Chemical

Historically, solvents such as oil of turpentine or chloroform have been used to soften gutta-percha, making it easier to remove by instrumentation with, for example, Hedstrom files. This technique is not advisable for removal of gutta-percha prior to post space preparation as there is no control over the depth of softening, which could disrupt the apical seal and allow leakage of the solvents into the periradicular tissues.

Thermal

This is a very easy way to remove a predetermined length of gutta-percha from a root canal. Using a preoperative radiograph, a correct size of System B tip (SybronEndo), chosen to bind at the desired post length, is marked with a rubber stop. The System B is set at 100°C and the tip inserted into the gutta-percha for 3 seconds. The heat is removed and apical pressure maintained for a further 10 seconds. A short burst of heat is then applied, and the tip rotated and removed, bringing the coronal mass of gutta-percha with it. A plugger is then applied vertically to compact the residual gutta-percha. The canals walls can be examined under magnification to ensure all traces of material have been removed from the root canal walls.

Mechanical

This technique is popular but if not carried out with care can weaken or result in perforation of the root (Figure 7.11). It is advisable therefore to use a non-end cutting bur such as Gates Glidden (Figure 7.12) or a Peeso reamer. Modern rotary nickel titanium retreatment instruments such as the ProTaper D series (Dentsply, Surrey, UK; Figure 7.12) make gutta-percha removal very easy. As with heat removal, the apical gutta-percha should be vertically compacted to ensure a good seal.

Creating the post space

The longer the post, the better the retention; however, as mentioned previously, at least 4–5 mm of gutta-percha must be present apical to the post. A space between the end of the post and the root canal filling should be avoided as this will compromise the seal and may predispose to leakage and possible failure of the root canal filling. An adequate width of post is required for strength and resistance to post fracture and this will vary with each post system; however, broad posts should be avoided as there is a high risk of lateral perforation and a greater chance of root fracture. It has been suggested that the ideal post diameter at its apical end should be no more than one-third of the root width at this level.



Figure 7.11 Inappropriate post space preparation in the upper right central incisor has nearly led to a distal perforation.



Figure 7.12 Gates Glidden burs (left) and ProTaper D series (right) used to remove gutta-percha prior to post space preparation. Note a rubber stop should be placed on the instrument at the desired post length.

When using the twist drills to create the post space it is important to start with the smallest size, gradually increasing to the optimum size to minimize heat build-up, reducing potential for damage to cells within the periodontal ligament. This technique will also keep dentine removal centred along the root canal and reduce the risk of lateral perforation.

Types of post

Metal posts

Metal posts are available in a variety of metals including stainless steel, titanium and titanium alloys, gold alloys, non-precious metals and gold plated brass. Stainless steel and brass posts can form corrosion products within the post space that can lead to discolouration of the root and compromise the aesthetics. Titanium posts do not corrode but fracture more easily and have a similar radio-opacity to gutta-percha, often making them difficult to differentiate on a radiograph. This can, on occasion, lead to a surprise finding when a decision is made to carry out root canal retreatment.

Posts are either parallel or tapered in design. Although parallel posts are more retentive, there is a greater risk of perforation of the apical root during preparation if a wide post is placed in a narrow canal.

Active posts

These involve cutting threads in the root canal dentine prior to placement and therefore tend to be more retentive than passive posts of a similar design. They are either self-threading or pre-tapped.

Self-threading posts have a shank that is slightly narrower than the prepared post space and a thread that is slightly wider. As the post is rotated into place it cuts its own counter-thread into the dentine. The procedure is quite unlike using a wood screw since dentine is hard and brittle. Careful placement is needed, using concepts more similar to metal working than wood, as there is potential to introduce stresses in the root dentine. Once the post hole has been prepared, it is tempting to coat the self-threading post with luting cement and rotate it into place in one stage; however, this introduces stresses into the dentine. Instead, the threaded post should be tried-in, cutting its own counter-thread in the dentine, and if retentive it is removed and reinserted into place coated with a cement lute. Used in this way the amount of stress introduced into the dentine can be reduced.

Self-threading posts can be either tapered or parallel in design. An example of a tapered, threaded post is the Dentatus screw (Dentatus) (Figure 7.13). This post can cause high stresses within the root as it is inserted, causing a wedging effect and possible root fracture. The Radix Anchor (Dentsply) is an example of a parallel, self-threading post. The threads are located at the coronal portion of the post only (Figure 7.13). Again the post should be placed initially without cement to cut the counter-thread. When it is reinserted to the full length of the prepared post space it should be derotated by a quarter turn to minimize stresses within the root dentine.

Pre-tapped posts have a higher frequency of threads around a parallel-sided shank. The post space is prepared and then the counter-thread is cut into the dentinal walls by using a thread cutter or pre-tapping device (Figure 7.14). This is designed to reduce the stresses created compared to that induced when self-threading posts are used in a one-stage technique. The higher frequency of threads and the lack of a vent to provide an escape route for excess cement can cause excessive stresses within the dentine. An example of this type of post is the Kurer K4 anchor system (Sabre Dental; Figure 7.13).

Although active posts allow for greater retention, the stresses they produce on the root dentine are not favourable, even when the pre-tapped varieties are used, and may result in root fracture. It would be sensible to reserve these posts for cases where retention is compromised such as short or curved roots.

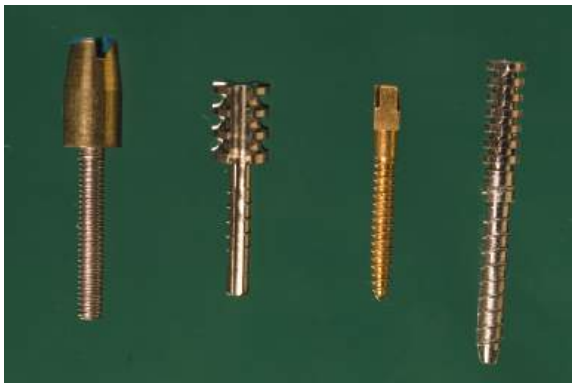


Figure 7.13 Examples of active (threaded) posts. Starting left with two parallel sided posts the Kurer K4 and Radix Anchor, then two tapered posts, the Dentatus screw and FlexiPost. The last has a split down its centre which allows the post to collapse when screwed into place.



Figure 7.14 Kurer K4 post drills and tapping device (right). From left to right: reamer to remove gutta-percha, post space drill, root facer (to produce a flat surface to seat the post down to) and the tapping device to cut the counter threads on the post hole wall. The actual post is seen in Figure 7.13.

Passive posts

These are either custom-made cast posts or prefabricated posts. Custom-made cast posts and cores can be made from type III, type IV gold alloy or base metal alloys. The latter are harder and may predispose the tooth to root fracture. The impression technique produces a smooth-sided, tapered post that matches the original taper of the root canal preparation. This is an ideal choice for an irregular canal such as that seen in an upper second premolar tooth, which is wide and oval bucco-palatally at its entrance and much narrower apically (Figure 7.15). To take an impression of an irregular post hole the wash material is syringed around the entrance to the post hole and the plastic impression post is pushed through the unset wash, dragging it into the post hole. Before the wash sets the loaded impression tray is seated (see Chapter 13).

Although the use of cast tapered posts decreases the risk of root perforation apically, the retention of the post is compromised. Parallel-sided posts give better retention than tapered posts and can be used provided the rules set out under post space preparation are adhered to. Serrated posts have negative recesses in them and give more retention than smooth-sided posts. It is possible to custom-make a parallel-sided serrated post by using a system such as the ParaPost XP (Coltène Whaledent, UK; Figure 7.16). For each size post there is a smooth plastic impression post (the blue one can be seen in Figure 7.15). A retentive 'mushroom' shape can be created in the impression post at the end which is embedded into the impression by melting it with a hot metal instrument such as a flat plastic. The serrated burn out plastic post is then sent with the impression to the laboratory for the technician to wax a core around and cast using the lost wax technique (see Chapter 9). The final post type within the parapost XP kit is the metal temporary post which can be used in conjunction with temporary crown materials and techniques (see Chapter 14).

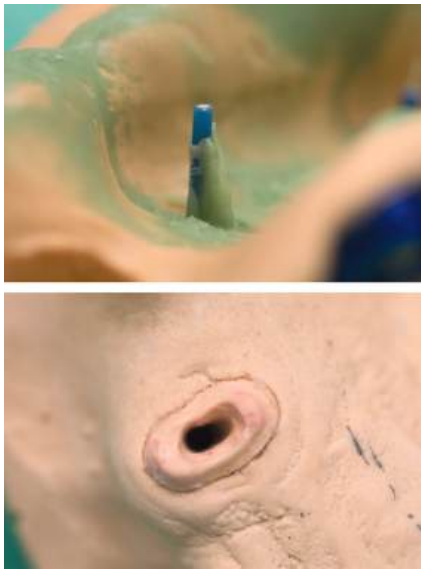


Figure 7.15 Impression taken for a cast post and core (top). A blue plastic impression post has been used with an addition cured silicone putty and wash impression material. The resultant die stone model can be seen below.

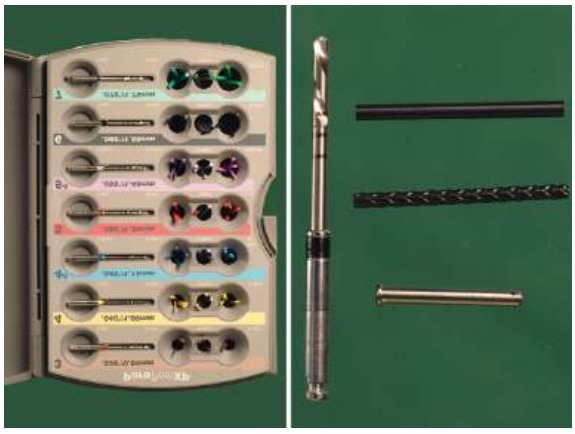


Figure 7.16 ParaPost XP system for making custom-made cast post and cores. On the right magnified image, from top to bottom is the plastic impression post, the serrated burnout post and the metal temporary post.

The use of custom-made cast posts and cores is more time consuming as additional clinical time and laboratory costs are incurred. It is preferable to fit the cast post and core and then take a further impression for the crown to reduce the chance of a compromised marginal fit (Figure 7.17). However, cast posts are preferred by many practitioners and so continue to have a use in operative dentistry.

Prefabricated passive posts

These can be either tapered or parallel in design. They have a vent that allows the escape of excess luting cement. Examples of posts from the ParaPost series (Coltène Whaledent, UK) have parallel sides and a diamond-shaped retentive feature (serrations) along part of (ParaPost XT) or along the entire length of (ParaPost XH) the post. The diamond-shaped serrations interconnect, effectively acting as a vent for excess luting cement (Figure 7.18). Once cemented into the post hole, the core is built up directly in the mouth with the chosen core material.

Non-metal aesthetic posts

These were introduced to improve aesthetics as metal posts can be visible through all-ceramic restorations and can cause the marginal gingival tissues to appear dark. Aesthetic clear, white or tooth-coloured posts can be made from quartz fibre, ceramic and zirconia.



Figure 7.17 Preparation for cast post and core and metal–ceramic crown (top left). The cast post and core (top right) is cemented (middle left) and a second impression taken for the metal–ceramic crown (bottom). A minimum preparation (resin retained) bridge was made at the same time to replace the lower left first premolar tooth.



Figure 7.18 Examples of prefabricated posts which are taken from the ParaPost series (Coltène Whaledent, UK). Left is the ParaPost XT which has an apical serrated pattern and a coronal 7 mm of threaded post. The ParaPost XH has the same serrated design along its entire length (not shown). The two quartz fibre posts are the Fiber White (middle) and Taper Lux (right). The red ring is for colour coding only and should be removed before the core is built up.

Fibre posts

Carbon fibre posts were introduced in the 1990s in an attempt to overcome the complication of root fracture that existed with metal posts. They were more flexible, having a similar stiffness to dentine, and were bonded to the post space using adhesive resin cements. These have since been replaced by tooth-coloured quartz, silica and glass fibre posts, which give superior aesthetics. The typical composition by weight is 42% fibre, 29% filler and 18% resin. The quartz fibres run longitudinally along the length of the post embedded in an epoxy resin (Figure 7.19). There are tapered and parallel designs available that are compatible with placement of a composite core; some even have retentive head designs to improve core retention (Figure 7.18).

Whilst the microflexure of these posts reduces the risk of failure due to vertical root fracture, in the authors' opinion they are best used when a ferrule can be achieved. If there is no coronal tooth tissue, the composite core can debond from the root face and the microflexure can lead to either post fracture or delamination of the post due to leakage of moisture (Figure 7.20).

When preparing the post space for a quartz fibre post it should be meticulously cleaned of traces of gutta-percha and sealer, especially sealer containing eugenol which could interfere with the resin luting cements advised for cementing these posts. The post space should be carefully dried prior to cementation using a paper point. Self-etching, resin luting cements are ideal for cementing quartz fibre posts as the reliable application of acid etch, rinsing, drying and application of primer and bond is difficult in the narrow confines of a post hole.

Should endodontic retreatment be required, quartz fibre posts can be removed by cutting along the post.

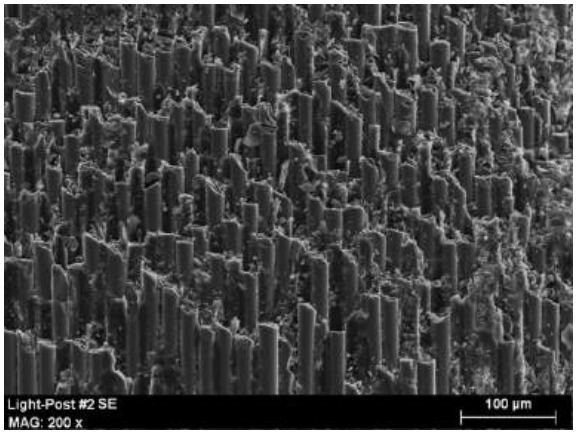


Figure 7.19 Micrograph of a quartz fibre post – Light-Post. The fibres can be seen running longitudinally along the length of the post embedded in an epoxy resin.
(Courtesy of RTD, France)

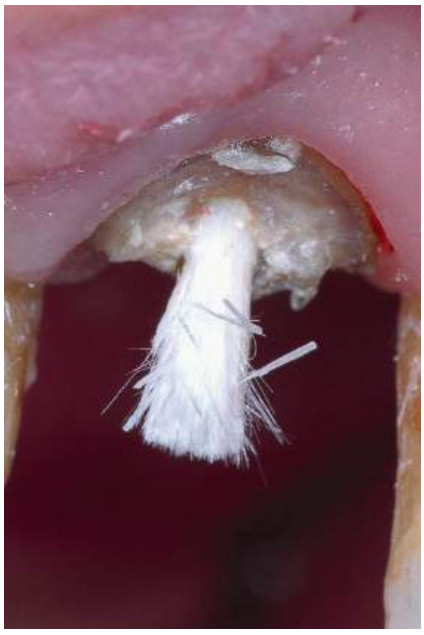


Figure 7.20 This tooth has no coronal tooth tissue present for a ferrule. The composite core has debonded and fluid microleakage has led to delamination of the post and loss of the crown. The make of post is unknown.

A pilot hole is cut using a pin drill followed by a non-end cutting rotary instrument such as a Peeso drill (Figure 7.21). Removing quartz fibre posts can be difficult because they are tooth coloured and in the depths of a post hole it is almost impossible to determine what is post and what is tooth tissue. Recherches Techniques Dentaires (RTD) has overcome this problem with their new post system, D.T. Light-Post Illusion. These posts have a colour added to them which is temperature sensitive; at body temperature they are clear/tooth coloured but when cooled with a three-in-one syringe, for example, the colour reappears, assisting in their removal if required (Figure 7.22).



Figure 7.21 Quartz fibre post retrieval kit provided by RTD. *(Courtesy of RTD, France)*

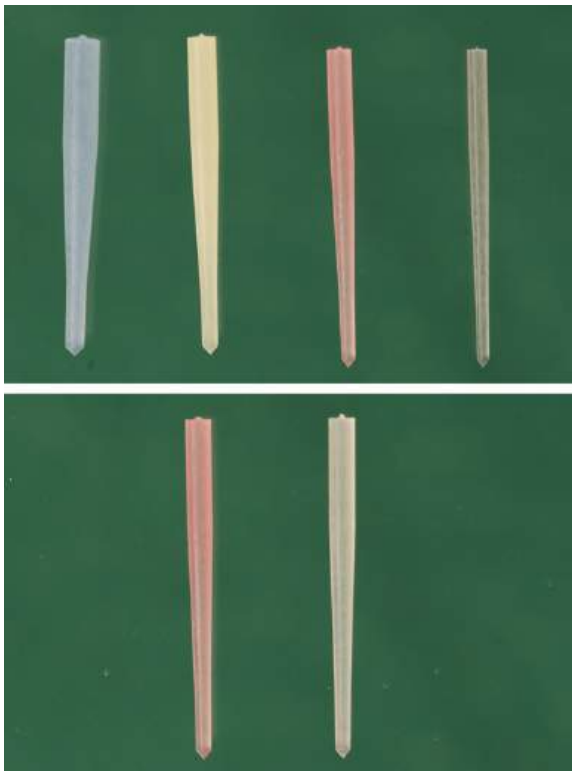


Figure 7.22 D.T. Light-Post Illusion. The colour-coded posts have double tapered configuration (top). When warmed to body temperature the colour disappears (bottom post on right).

Fibre posts and weakened roots

In cases of weaken roots such as immature teeth or those compromised by caries or operative procedure, the internal surface of the root can be strengthened using composite bonded to the root dentine, prior to placement of a quartz fibre post. In order to ensure the composite material is cured along its length, a clear plastic sprue (Luminex, Dentatus) is placed through the middle of the composite and then removed following curing. This leaves a ready-made channel that just requires refinement before bonding the post (Figure 7.23).

Ceramic and zirconium posts

Although these posts are tooth coloured they have several disadvantages in that they are weaker than metal posts and therefore have to be made broader, resulting in a wider post-space preparation that may result in perforation. Zirconium posts cannot be etched, making retention of a composite core very difficult. Both are extremely

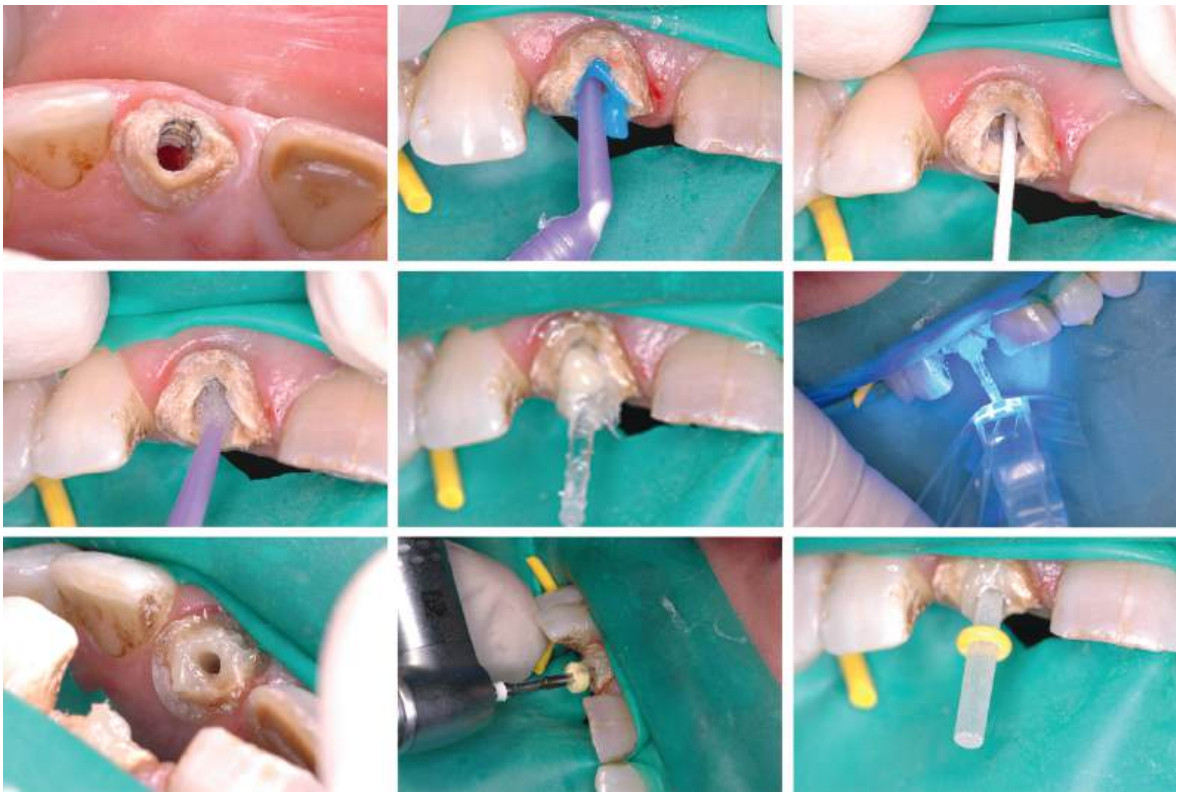


Figure 7.23 From left to right and top to bottom. Severely compromised root due to poor post hole preparation and near perforation. Acid etch applied and agitated in post hole with microbrush for 10 seconds. After rinsing thoroughly the post hole is dried with a large paper point. Bond is then applied with a microbrush. The post hole is filled with composite and the petroleum jelly-covered Luminex post inserted through the composite and light cured. The post is removed, post hole modified with a suitable twist drill and the chosen quartz fibre post cemented with a resin luting cement. The post length can now be reduced if required and a composite core built up.

difficult to remove and often this can only be done using a diamond high speed bur; however, this is very tedious and carries with it a high risk of the bur skating off the post into the dentine and perforating. It is best to avoid this type of post.

POSTERIOR TEETH

Endondontically treated posterior teeth should be restored with a cuspal coverage restoration to protect the remaining tooth structure and prevent failure due to fracture or coronal leakage. The type of restoration depends on the amount of remaining tooth structure and the occlusal forces on the tooth. Placing posts in premolar and molar teeth can cause complications such as a strip or lateral perforation and can weaken the root, predisposing to root fracture. As retention can be gained from the undercut shape of the pulp chamber in combination with mechanical and adhesive retention, posts are generally not required to retain the core material and are therefore unnecessary.

Nayyar described a core that used the pulp chamber and the coronal 2–4 mm of the root canals for retention. The roots of posterior teeth usually diverge from the pulp chamber and removing 2–4 mm of root filling material and packing core material into the space provided affords excellent retention. This technique therefore prevented the need for the placement of pins and posts in posterior teeth. More recent research has shown that removing gutta-percha at the entrance to the root canals is no longer necessary. If the access cavity for the endodontic treatment has been cut correctly, the pulp chamber itself should be undercut. The cross-sectional area of core material within the pulp chamber is also adequate for acceptable strength



Figure 7.24 Modification of a Nayyar core. The pulp chamber is utilized for retention; however, no gutta-percha has been removed from the entrance to the root canal. The core has been built up in composite.

of any of the core materials in use. If amalgam is used it can be bonded to the tooth structure for added retention and to improve the coronal seal; however, the evidence for this is weak. Composite can also be used as a core material but should not be placed into the root canals as this would be difficult to remove, in that it would be difficult to differentiate it from tooth tissue, should root canal retreatment be required (Figure 7.24).

SUMMARY FOR PLACEMENT OF POST AND CORE MATERIALS

Retention and resistance

Retention is the ability to resist vertical displacement and is influenced by the post's length, diameter, taper, whether active or passive, and the type of luting cement used. Parallel posts are more retentive than tapered posts but have a higher risk of perforation with increasing diameter. Active posts are more retentive than passive posts but if placed with too much torque can result in root fracture. The post should be as long as possible but maintain an apical seal of 4–5 mm gutta-percha. For success, the length of the post should be equal to if not greater than the clinical crown height.

Resistance is the ability of a post to withstand lateral and rotational forces. This is determined by the amount of remaining coronal tooth structure, the length of post, the presence of a ferrule, and whether or not an anti-rotational key has been cut in the root surface. The last is only required with cylindrical posts when there is no coronal tooth tissue for a ferrule and should be cut on the palatal/lingual aspect of the post hole to a depth of about 2–3 mm.

Ferrule effect

The length of the post and creation of a ferrule are the two most important determinants for a successful post crown. The presence of a ferrule is an important feature that will increase the longevity of the restoration. It refers to the presence of the coronal tooth structure that the crown will encompass and should be at least 2 mm in height. This will improve the fracture resistance of the tooth and reduce the risk of vertical root fracture, especially when metal posts are used. The band of extracoronal material that covers this tooth structure is the ferrule and is usually provided by the crown that is placed over the core (Figure 7.25).

Posterior teeth

Avoid the placement of posts in posterior teeth whenever possible as preparation for these can easily result in a strip perforation. Rely instead on retention from placement of the core material within the pulp chamber, together with the aid of bonding.

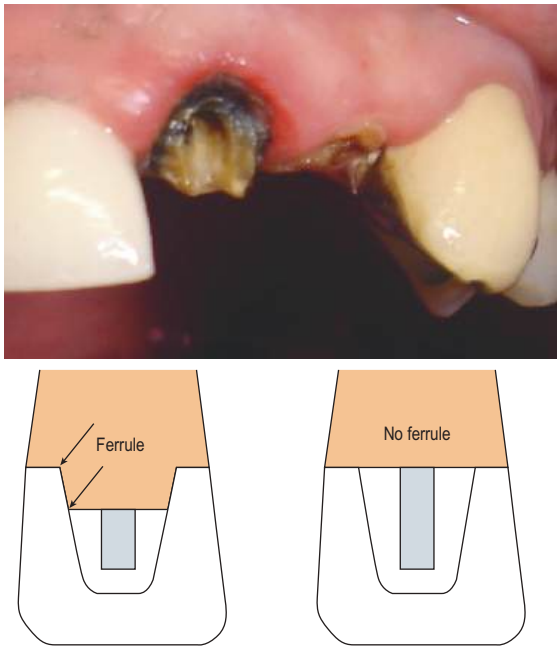


Figure 7.25 Two examples of teeth that require posts to retain cores, the left with a 2–3 mm ferrule mesio-distally and the right with no coronal tooth tissue and no ferrule. The tooth with the ferrule has a much better prognosis than the one without.

FURTHER READING

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